

# Davis Rattनाविजै

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## ABOUT ME

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Mechanical engineering student interested in supply chain and operations. My experience spans 1,500+ hours in Formula SAE across technical design, fabrication, testing, and procurement, 200+ hours in manufacturing, and 600+ hours in customer service.

## EDUCATION

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**Cornell University, Duffield College of Engineering,** Ithaca, New York

Expected May 2028

Bachelor of Science in Mechanical Engineering GPA: 3.73

## EXPERIENCE

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### Machine Shop Staff

Feb 2026 - Present

*Manufacturing Learning Studio, Cornell University, Ithaca, NY*

- Operate manual mills, lathes, 3D scanners, and Haas CNC equipment for manufacturing, inspection, and maintenance
- Train engineering students in machine operation, tool selection, tolerance analysis, and efficient machining practices
- Develop technical manuals, safety documentation, and instructional materials to standardize equipment operation

### Suspension Lead

Jun 2026 - Present

### Suspension Engineer and Purchasing Coordinator

Feb 2025 - Jun 2026

*Cornell Formula SAE Racing, Cornell University, Ithaca, NY*

- Work with 70+ engineers across 10 subteams to design and build an electric race car for the annual Formula SAE competition
- Lead a 5 person team through design, sourcing, manufacturing, assembly, and validation of the suspension system
- Responsible for vehicle dynamics simulations and parameterized models, evaluating design changes while coordinating cross-functional dependencies across suspension, chassis, steering, tires, and aerodynamics
- Designed a custom welding fixture for the suspension links, targeting tolerance control, and manufacturability
- Oversaw \$200,000+ in procurement operations; handling invoicing, purchasing, receiving of in orders across all sub-teams
- Scaled and improved the purchasing process by expanding the team from 2 to 7 members, standardizing communication and workflows, and strengthening accountability for order tracking, records, and bookkeeping
- Wrote a 200+ page supplier RFQ for the FSAE Business Case, translating our engineering requirements into BOMs, manufacturing specifications, quality standards, and delivery timelines

### Undergraduate Teaching Assistant

Dec 2025 - May 2025

*Department of Physics, Cornell University, Ithaca, NY*

- Lead weekly recitation for 20 students in Physics II, guiding rigorous multi-step derivations and analytical problem solving
- Strengthen physics knowledge and teaching skills through a concurrent course in evidence-based physics instruction

### Seasonal Teller

May 2025 - Aug 2025

*Middlesex Savings Bank, Wayland, MA*

- Provided in-branch and phone customer service to hundreds of clients, including resolving conflicts and frustrations
- Processed thousands of customer transactions, including deposits, withdrawals, loan payments, check cashing, and CDs
- Ensured day-to-day compliance with identity verification, dual control, audits, fraud, and documentation procedures

## ADDITIONAL EXPERIENCE

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**Team Lead, Mathematical Modeling Team, Newton North High School,** Newton, MA

Nov 2021 - May 2024

- Led a four-person team to three finalist placements and a 3rd-place \$10K scholarship at seven international competitions
- Analyzed categorical, time-series, and geospatial data; built quantitative models and forecasts using Python, MATLAB, and Excel to evaluate scenarios, identify trends, and develop data-driven recommendations

**Captain, Intramural Soccer Team, Cornell University**

Mar 2025 - Present

**Tour Guide, Office of Visitor Relations, Cornell University**

May 2026 - Sep 2026

## SKILLS & INTERESTS

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**Technical Skills:** Microsoft Excel, Python, MATLAB, Autodesk Inventor, Fusion 360, Manual & CNC Machining

**Professional Skills:** Customer Service, Cross-Functional Collaboration, Project Management, Attention to Detail

**Languages:** Mandarin (Conversational), Japanese (Limited Proficiency)

**Interests:** Soccer, Real Madrid FC, Formula 1, Entrepreneurship/Product Development, Stocks, Biking, Skiing, Dragon Boating